Port Cyber Security

Shifting from Risk toResilience

TEAM A-6

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Problem & Gap

Problem :

- Seaports are becoming increasingly digital and interconnected.
- Cyber incidents can disrupt port operations and global supply chains.
- A cyberattack on one critical port can have cross-border impacts.

Key Gaps :

- Different levels of cybersecurity maturity across ports.
- Limited threat intelligence sharing and incident-response coordination.
- Supply-chain cybersecurity remains a challenge.
- Differences in policies and regulations across jurisdictions.
- Limited coordination between governments, port authorities, and private-sector stakeholders.

Core challenge :

How can ports move from managing cyber risk to building coordinated, scalable, and sustainable cyber resilience?



A GLOBALLY COORDINATED RESILIENCE MODEL

The Solution

Build one shared model connecting policy, law and technology, while allowing every port to progress from its current maturity level.

1 Policy & Governance

Cross-border cooperation framework

Clear ownership through an international public-private partnership

Sustainable public and private funding

2 Legal Alignment

Trusted information-sharing rules

Globally applicable principles translated from IMO guidance and NIS2

Proportionate requirements that avoid fragmentation

3 Technology & AI

Port Cyber Passport for continuous resilience assessment

Port Cyber Emergency Room for detection, containment and coordination

Secure intelligence sharing with human oversight

CONTINUOUS READINESS AND COORDINATED RESPONSE

The Solution: How It Works

1. **ASSESS** Cyber Passport maps critical assets, vulnerabilities, supply-chain exposure and recovery readiness.
2. **IMPROVE** AI identifies gaps and recommends priority improvements suited to each port's maturity.
3. **DETECT** AI-assisted monitoring identifies anomalies and classifies significant threats.
4. **RESPOND** Cyber Emergency Room recommends containment actions, with human approval for safety-critical decisions.
5. **SHARE & LEARN** Trusted ports and authorities receive secure, anonymised intelligence and improve together.



TRUST, ACCOUNTABILITY AND SUSTAINABLE COOPERATION

Governance & Legal

Policy Framework

• Common global principles for port cyber resilience

• Cross-border protocols for prevention, notification, assistance and recovery

• National implementation aligned to local risk and capacity

Ownership & Funding

• International steering body supported by IMO, UN, World Bank and national governments

• Operational delivery through public-private partnerships

• Tiered membership, public seed funding and private-sector contributions

Legal & Regulatory Alignment

• Trusted sharing agreements with confidentiality and liability safeguards

• Map IMO guidance and NIS2 into common outcomes

• Mutual recognition and proportionate controls to avoid duplication



THE DIGITAL ENGINE OF PORT RESILIENCE

Technology & AI

Port Cyber Passport

A continuously updated view of each port's critical assets, vulnerabilities, supply-chain exposure and recovery readiness. AI analyses the information to identify gaps and recommend priority improvements.

Port Cyber Emergency Room

Activates when a significant threat is detected. AI-assisted monitoring identifies anomalies, classifies the incident and recommends containment actions. Trusted ports and authorities receive anonymized threat intelligence through a secure global platform.

Human-led and inclusive: AI supports experts rather than replacing them. Maturity-based tiers let smaller ports start with monitoring and training, while advanced ports use predictive analytics and automated response playbooks.

Expected Impact

FROM RISK TO RESILIENCE

Faster detection

Shared monitoring and AI-assisted anomaly identification.

Stronger coordination

Common playbooks and cross-border response support.

Trusted intelligence

Secure, anonymized learning across the ecosystem.

Inclusive progress

Maturity tiers enable participation by ports of every size.





Thank You!

From isolated risk management to coordinated, scalable and sustainable cyber resilience.